

any loop we required 3 things

- initialization : where you want start
- increment or decrement : how much you want increment or decrement
- condition to stop : At which point you want

```
In [ ]: for i in range(1,10,2)

range(1,10,2) start=1    increment/step=2    stop=10-1=9
```

```
In [ ]: # wrapper

# intializaion

# condition

# increment/decrement

#####

i=0 # intialziation

while <condition>:

    i=i+1 # increment
```

```
In [ ]: for i in range(5):
        print(i)

range(5) ===== > 3 seperate lines in while loop

i=0
while <condition>:
    print(i)
    i=i+1

# valid conditions based on i
# i<5  0<5 True   valid
# i>=0   valid
# i==0   valid
# i<1000
```

```
In [ ]: # I want to print 5 numbers
i=0
while i<5:    # i<=5
    print(i)
    i=i+1

# i=0    0<6    T    print(0)    i=0+1=1
# i=1    1<6    T    print(1)    i=1+1=2
# i=4    4<6    T    print(4)    i=4+1=5
# i=5    5<6    T    print(5)    i=5+1=6
# i=6    6<6    F    hard stop
```

```
In [ ]: # the below code iterate 5 times
# what I want to these 5 times

i=0
while i<5:
    print(i)
    i=i+1
```

```
In [ ]: #####Do not run below code  infinte loop#####
i=0
while i>=0:
    print(i)

# i=0    0>=0 T    print(0) 0
```

```
In [ ]: ##### Infinite Loop #####
i=0
while i>=0:
    print(i)
    i=i+1

# i=0    0>=0    print(0)    i=0+1=1
# i=1    1>=0    p(1)        i=1+1=2
# i=2    2>=0    p(2)        i=2+1=3
```

```
In [1]: i=0
while i>=0:
    print(i)
    i=i+1
    if i==10:
        break
```

```
0
1
2
3
4
5
6
7
8
9
```

```
In [3]: i=0
while i<5: # good condition
    print(i)
    i=i+1
```

```
#####
i=0
while i>=0: # Bad condition
    print(i)
    i=i+1
    if i==6:
        break
```

```
0
1
2
3
4
0
1
2
3
4
5
```

```
In [4]: i=0
while True:
    print(i)
    i=i+1
    # to stop the Loop
    if i==5:
        break
```

```
0
1
2
3
4
```

```

In [ ]: # printing first 5 numbers 0 to 4
i=0
while i<5: # good condition
    print(i)
    i=i+1

#####

i=0
while i>=0: # Bad condition
    print(i)
    i=i+1
    if i==6:
        break

#####

i=0
while True:
    print(i)
    i=i+1
    # to stop the loop
    if i==5:
        break

```

```

In [ ]: # print -1 to -5 using for loop
for i in range(-1,-6,-1):
    print(i)
# print -1 to -5 using while loop
i=-1
while i>-6:
    print(i)
    i=i-1

```

```

In [6]: # print -5 to -1 using for loop
for i in range(-5,0,1):
    print(i)

# print -5 to -1 using while loop

i=-5
while i<0:
    print(i)
    i=i+1

```

```

-5
-4
-3
-2
-1
-5
-4
-3
-2
-1

```

```
In [8]: # print 5 to 1 using for loop
for i in range(5,0,-1):
    print(i,end=' ')
# print 5 to 1 using while loop
i=5
while i>0:
    print(i,end=' ')
    i=i-1
```

5 4 3 2 1 5 4 3 2 1

```
In [9]: # wap ask the user print square of the numbers from 1 to 5
# do it for loop
for i in range(1,6):
    print(f"the square of {i} is {i*i}")
# do it with while loop
```

the square of 1 is 1
the square of 2 is 4
the square of 3 is 9
the square of 4 is 16
the square of 5 is 25

```
In [11]: i=1
while i<6:
    print(f"the square of {i} is {i*i}")
    i=i+1
```

the square of 1 is 1
the square of 2 is 4
the square of 3 is 9
the square of 4 is 16
the square of 5 is 25

```
In [ ]: for i in range(1,6):
        print(f"the square of {i} is {i*i}")

i=1
while i<6:
    print(f"the square of {i} is {i*i}")
    i=i+1
```

```
In [12]: # wap ask the user generate a random number between 1 to 20
# and print square of the number

# do that above task 4 times

# using for Loop
import random
for i in range(4):
    num=random.randint(1,20)
    print(f"the square of {num} is {num*num}")
```

```
the square of 17 is 289
the square of 10 is 100
the square of 17 is 289
the square of 1 is 1
```

```
In [13]: # using while Loop
i=0
while i<4:
    num=random.randint(1,20)
    print(f"the square of {num} is {num*num}")
    i=i+1
```

```
the square of 2 is 4
the square of 12 is 144
the square of 7 is 49
the square of 15 is 225
```

```
In [ ]: # WAP ask the user get a random number 20,70
# get even count and odd count
# Iterate that 5 times
# For Loop and while Loop
```

```
In [ ]: # Quick assignement
# For Loop ===== while Loop
```

```
In [ ]:
```